

Data Source Integration – Proficient in connecting, importing, and transforming data from a wide range of sources, including:

- ❖ **Relational Databases:** SQL Server, MySQL, PostgreSQL, Oracle.
- ❖ **Cloud Platforms:** Microsoft Azure (Synapse, Data Lake, Blob Storage), Google BigQuery, AWS services.
- ❖ **APIs & Web Services:** REST APIs, JSON, XML, OData feeds for real-time and batch data ingestion.
- ❖ **NoSQL Databases:** MongoDB, Cosmos DB.
- ❖ **Scripting Integrations:** Python and R scripts for custom ETL, API calls, and advanced data transformations.
- ❖ **File-Based Sources:** Excel, CSV, Parquet, JSON, XML.
- ❖ **Direct & Live Connections:** Expertise in DirectQuery, Import, and Hybrid connection modes for optimized performance.

Step-by-step integration workflow for connecting each type of data source in Power BI — this will help you cover APIs, SQL, Azure, BigQuery, MongoDB, Python scripts, and more.

1 General Workflow for Any Data Source

1. **Open Power BI Desktop.**
2. **Home → Get Data.**
3. Choose the **specific data connector** from the list.
4. **Authenticate** (enter credentials, API keys, or connection strings).
5. Use **Navigator** to select required tables/data objects.
6. Transform data in **Power Query** (clean, rename, merge, filter).
7. **Load** into the Power BI model (Import / DirectQuery / Hybrid).
8. Build your **semantic model** with relationships, measures, and hierarchies.

2 Step-by-Step for Each Source

A. SQL Database (SQL Server / MySQL / PostgreSQL / Oracle)

- ❖ **Home → Get Data → SQL Server** (or relevant DB connector).
- ❖ Enter **Server Name** and **Database Name**.
- ❖ Choose **Import** or **DirectQuery**.
- ❖ Authenticate with **Windows / Database credentials**.
- ❖ Select tables or write a **SQL query** for filtered data.

- ❖ Load into Power BI.

B. Microsoft Azure

Azure SQL Database / Synapse

- ❖ Get Data → Azure → Azure SQL Database / Azure Synapse Analytics.
- ❖ Provide **Server name** and database, choose **connection mode**.

Azure Data Lake / Blob Storage

- ❖ Get Data → Azure → Azure Data Lake Storage Gen2 / Blob Storage.
- ❖ Provide **URL & credentials**.
- ❖ Navigate to desired container/file.

C. Google BigQuery

Home → Get Data → More → Online Services → Google BigQuery.

Sign in with **Google account**.

Choose **Project** → **Dataset** → **Table**.

Decide on Import or DirectQuery.

Load and model.

D. MongoDB (via ODBC or BI Connector)

- ❖ Install **MongoDB BI Connector** or ODBC driver.
- ❖ In Power BI → Get Data → ODBC.
- ❖ Select the MongoDB DSN you created.
- ❖ Authenticate, select collections, and load.

E. REST APIs / Web Services

- ❖ **Home** → **Get Data** → **Web**.
- ❖ Enter **API URL** (e.g., `https://api.example.com/data`).
- ❖ If needed, add **API key** or **OAuth** in authentication.
- ❖ Parse JSON/XML in Power Query using **Transform** → **Parse** options.
- ❖ Clean and load.

F. Python Script Integration

- ❖ Install **Python** on your machine.
- ❖ In Power BI → **File** → **Options** → **Python scripting**, set Python home directory.
- ❖ Home → Get Data → **Python script**.
- ❖ Paste your script (e.g., API calls, web scraping, data processing).

- ❖ Power BI executes it and shows the returned dataframe.

G. File-Based Sources (Excel / CSV / JSON / Parquet)

- ❖ Home → Get Data → **Excel / Text/CSV / JSON / Parquet**.
- ❖ Browse and select the file.
- ❖ Preview and transform.
- ❖ Load to the model.

3 Best Practices for Integration

- ❖ Always rename fields and set correct data types in Power Query.
- ❖ Use Parameters for dynamic server names, file paths, or API URLs.
- ❖ For large datasets, prefer DirectQuery or Incremental Refresh.
- ❖ Document data source credentials and gateway requirements.

What-If Analysis for Mathematical Simulation Model Development

1. Concept

- ❖ What-If Analysis is a scenario-testing technique where you change one or more input variables to see how they affect the model's output.
- ❖ In mathematical simulation models, it is used to test real-world “what happens if...” situations without risking actual operations.

2. Core Idea in Simulation

- ❖ Inputs → Changeable parameters (e.g., demand rate, production cost, interest rate).
- ❖ Model → Mathematical formulas or algorithms (e.g., linear equations, regression, forecasting models).
- ❖ Outputs → KPIs or results (e.g., profit, cost, time, efficiency).

3. Steps

- ❖ Define the Variables Identify parameters that can change (e.g., growth rate, temperature, machine speed).
- ❖ Build the Mathematical Model Represent the system using equations, statistical models, or algorithms.
- ❖ Create Scenarios Assign different values to the variables (Best case, Worst case, Expected).
- ❖ Run Simulations Observe how output metrics change under each scenario.
- ❖ Analyze Sensitivity Identify which variables have the biggest impact.

4. In Power BI

- ❖ Use What-If Parameters to dynamically adjust values.
- ❖ Connect parameters to DAX measures.
- ❖ Display results in real-time using visuals (line charts, KPI cards, tables).

5. Example

Scenario: Predict sales revenue based on a variable *discount rate*.

- ❖ Input Variable: Discount Rate (%)
- ❖ Formula: $\text{Revenue} = \text{Units Sold} \times \text{Price} \times (1 - \text{Discount Rate})$
- ❖ Simulation: Adjust discount rate from 0% to 50% in 5% steps.
- ❖ Observation: Identify the discount rate that maximizes revenue.

6. Benefits

- ❖ Supports decision-making under uncertainty.
- ❖ Reduces business risk by testing outcomes before implementation.
- ❖ Improves strategic planning through evidence-based forecasting.

Q&A Query Development for On-Demand Analysis – Basics

1. Concept

- ❖ **Power BI Q&A** allows users to **ask questions in natural language** and get instant answers in the form of visuals.
- ❖ It is used for **on-demand analysis** — meaning you can explore data without pre-built reports.

- ❖ Example: Typing "*Total sales by region for 2024*" generates a chart instantly.

2. How It Works

- ❖ **Data Model** → Must have well-named fields, relationships, and measures.
- ❖ **Q&A Engine** → Uses AI to interpret user queries.
- ❖ **Visual Generation** → Automatically picks the best chart for the query.

3. Steps to Develop Q&A Queries

1. Prepare the Data Model

- ❖ Use **descriptive names** for tables & columns.
- ❖ Create DAX measures for KPIs (e.g., `Total Sales`, `Profit Margin`).
- ❖ Set correct **data types** and categories (e.g., Dates, Geography).

2. Enable Q&A in Power BI

- ❖ In **Power BI Desktop**, turn on Q&A in report settings.

3. Teach Q&A with Synonyms

- ❖ Add **synonyms** in the *Linguistic Schema* so Q&A understands different terms (e.g., "Revenue" = "Sales").

4. Build & Test Questions

Try questions like:

- "Sales trend by month"
- "Top 5 products by profit"
- "Customer count in 2023"

5. Deploy for End-Users

Publish the report and allow users to type queries directly in the Q&A visual.

4. Benefits for On-Demand Analysis

- ❖ No need for pre-built filters or dashboards for every question.
- ❖ Instant insights in real-time.
- ❖ Empowers non-technical users to analyze data independently.

1. What is Microsoft Fabric?

- ❖ **Microsoft Fabric** is an **all-in-one analytics platform** from Microsoft that unifies data engineering, data integration, data science, real-time analytics, and business intelligence into a single environment.
- ❖ Think of it as **Power BI + Azure Synapse + Data Factory + Data Lake + Real-time streaming** all combined in one SaaS service.

2. Key Goals

- ❖ Break down **data silos** (different teams/tools for ingestion, storage, analytics).
- ❖ Provide **end-to-end analytics** — from raw data ingestion to business-ready dashboards.
- ❖ Make analytics **accessible** for both **business users** (via Power BI) and **technical teams** (via SQL, Python, Spark, and notebooks).

3. Core Components (Experiences)

Fabric has **seven main experiences** inside a single platform:

Experience	Purpose
Data Factory	Ingest and transform data from 150+ connectors (like Azure Data Factory).
Data Engineering	Create, manage, and process big data using Lakehouses and Apache Spark .

Experience	Purpose
Data Science	Build, train, and deploy ML models using Python/R and integrate with Azure ML.
Real-Time Analytics	Ingest and analyze streaming data for IoT, telemetry, and event-based use cases.
Data Warehouse	Store structured data in a SQL-based warehouse optimized for analytics.
Power BI	Create dashboards and reports directly within Fabric.
Data Activator	Automate actions based on data events (trigger alerts, workflows).

4. OneLake – The Foundation

- ❖ **OneLake** is the **single, unified data lake** for all Fabric workloads.
- ❖ Works like **OneDrive for data** — all items (lakehouses, warehouses, shortcuts) share the same underlying storage.
- ❖ **Delta Lake format** for open, optimized analytics.

5. Key Features

- ❖ **Unified Workspace** – All workloads in the same workspace.
- ❖ **No need for multiple tools** – Everything from ingestion to visualization in one UI.
- ❖ **Shortcuts** – Reference data without copying it.
- ❖ **Security** – Row-level, column-level, and object-level security.
- ❖ **Governance** – Purview integration for data cataloging and lineage tracking.

6. How It Connects to Power BI

- ❖ Fabric is **natively integrated** into Power BI.
- ❖ Power BI datasets can directly query Fabric **Lakehouses** and **Warehouses**.
- ❖ Real-time dashboards can be built directly on streaming data ingested in Fabric.

7. Example Use Case Flow

- ❖ **Ingest** – Pull sales, IoT, and social media data via **Data Factory**.

- ❖ **Store & Process** – Keep raw data in **OneLake** and clean/process using **Data Engineering (Spark)**.
- ❖ **Analyze in Real-Time** – Use **Real-Time Analytics** to detect anomalies in streaming IoT data.
- ❖ **Model & Predict** – Train a demand forecasting model in **Data Science**.
- ❖ **Visualize** – Build an executive sales & operations dashboard in **Power BI**.
- ❖ **Trigger Actions** – Use **Data Activator** to send alerts to managers when KPIs drop.

8. Why It's Important

- ❖ **End-to-end**: You don't jump between Azure Synapse, Data Factory, and Power BI.
- ❖ **Time saving**: Reduces ETL/ELT complexity.
- ❖ **Scalable**: Can handle small data marts and petabyte-scale analytics.
- ❖ **Future-proof**: Designed for AI-driven analytics and automation.

The step-by-step process to get started:

Prerequisites

- ❖ A **Microsoft Power BI account** (Power BI Pro or Premium Per User recommended).
- ❖ If you're in an **organization**, Fabric features may need to be **enabled by your Microsoft 365 / Power BI admin**.
- ❖ A **supported browser** (Edge, Chrome, or Firefox recommended).

Enable Microsoft Fabric

1. **Sign in** to Power BI Service.
2. Click on your **Profile icon** (top-right corner).
3. Select **Preview features**.
4. **Turn on “Microsoft Fabric”**.
5. Click **Save**.
6. **Refresh your browser**.

Activate Fabric in Admin Portal (*Org-wide step*)

(Only needed if you’re an admin or your admin hasn’t enabled it yet)

- ❖ Go to **Power BI Admin Portal** → **Tenant settings**.
- ❖ Find the **Microsoft Fabric** section.
- ❖ Enable **“Users can try Fabric”**.
- ❖ Save changes.

Access Fabric Features

After enabling, when you **create a new workspace** in Power BI, you’ll see **Fabric workloads**:

- ❖ **Data Factory**
- ❖ **Data Engineering**
- ❖ **Data Warehouse**
- ❖ **Real-Time Analytics**
- ❖ **Data Science**
- ❖ **Data Activator**

You can also explore **OneLake** from the left menu.

Start a Fabric Trial (if needed)

- ❖ If you don't have a Premium capacity, click **Start trial** when prompted in any Fabric experience.
- ❖ This activates a **60-day free Fabric trial** with full features.

Verify Installation

In Power BI Service, create a **Lakehouse** or **Warehouse** — if you can create them, **Fabric is active**.

Note: There's no local setup — all Fabric work is done in the **cloud** via the Power BI web portal, but you can still connect from **Power BI Desktop**, **SQL tools**, or **Notebooks**.